

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 9 - Summative Assessment 2

Summative Assessment (SA-2)

Class: Class 9

Assessment Type: Summative Assessment (SA-2)

Total Marks: 40

Duration: 2 hours

Topics Covered: Algorithms, Cyber Security, Emerging Technologies, Computational Thinking

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Part A: MCQ (10 x 1 = 10 marks)

- a. What is the time complexity of bubble sort in the worst case?
 - b. $O(n)$
 - c. $O(n \log n)$
 - d. $O(n^2)$
 - e. $O(\log n)$
- b. Which of the following is NOT a characteristic of a good algorithm?
 - c. Finiteness
 - d. Definiteness
 - e. Ambiguity
 - f. Effectiveness
- c. What does phishing refer to?
 - d. A type of malware
 - e. A social engineering attack to steal credentials
 - f. A hardware malfunction
 - g. A network protocol
- d. Which technology is the foundation of Bitcoin?
 - e. Cloud Computing
 - f. Artificial Intelligence
 - g. Blockchain
 - h. Big Data
- e. What is the primary purpose of a firewall?
 - f. To speed up internet
 - g. To filter network traffic and block threats
 - h. To store data
 - i. To compile programs
- f. In a flowchart, which shape represents an input/output operation?

- g. Rectangle
- h. Diamond
- i. Parallelogram
- j. Oval
- g. Which of the following is an example of **phishing**?
 - h. A virus infecting a computer
 - i. An email asking for password pretending to be from a bank
 - j. A hardware crash
 - k. A software update
- h. What is **Machine Learning**?
 - i. A type of hardware
 - j. A subset of AI that enables systems to learn from data
 - k. A programming language
 - l. A database system
- i. Which cryptographic technique uses the same key for encryption and decryption?
 - j. Asymmetric encryption
 - k. Symmetric encryption
 - l. Hashing
- m. Digital signature
 - j. What is the full form of IoT?
 - k. Internet of Technologies
 - l. Internet of Things
- m. Integration of Things
- n. Interface of Technology

Part B: Short Answer (8 x 2 = 16 marks)

- a. Write an algorithm (in pseudocode) to find the second smallest element in an unsorted list. Analyse its time complexity.
- b. Differentiate between **encryption** and **decryption**. Give one real-world example of each.
- c. Explain three types of **malware** with their characteristics and one prevention method each.
- d. What is **Big Data**? Explain the 5 V's of Big Data with one sentence each.
- e. Write a Python program to implement **selection sort** and trace it on [64, 25, 12, 22, 11].
- f. Explain **two-factor authentication**. Why is it more secure than a single password?
- g. What is **cloud computing**? Differentiate between **public cloud**, **private cloud**, and **hybrid cloud** with one example each.

- h. Write the pseudocode for a **binary search** algorithm. Trace it on [2, 5, 8, 12, 16, 23, 38, 56, 72, 91] to find 23.

Part C: Long Answer (4 x 3 = 12 marks)

- a. Write a complete Python program that implements **Merge Sort**. Trace the algorithm on [38, 27, 43, 3, 9, 82, 10] showing each divide and merge step. Analyse the time complexity for best, average, and worst cases.
- b. A school wants to implement a **cyber safety policy**. Write:
- Five common cyber threats students may face
 - Detailed prevention measures for each threat
 - A step-by-step incident response plan if a student's account is compromised
 - How the school can use technology to monitor and prevent cyberbullying
- c. Explain the following emerging technologies with detailed examples:
- **Artificial Intelligence**: How it works, two applications in education, and ethical concerns
 - **Internet of Things (IoT)**: Components, how sensors work, and one smart home application
 - **Blockchain**: Structure, how transactions are verified, and one application beyond cryptocurrency
- d. Design a **Sorting Algorithm Visualiser** project concept. Write:
- The problem statement and objectives
 - A Python program structure using dictionaries and lists to simulate sorting steps
 - How bubble sort, insertion sort, and selection sort would be implemented
 - How a web interface could display the sorting process with JavaScript animations

Part D: Application Based (2 x 1 = 2 marks)

- a. A social media company wants to protect user data. Explain how:
- Encryption can secure messages in transit
 - Firewalls and intrusion detection systems protect servers
 - Strong password policies and 2FA protect individual accounts
 - What regulations must the company follow regarding data privacy
- b. A healthcare facility wants to digitise patient records. Analyse how:
- Cloud computing can provide scalable storage for medical data
 - IoT devices can monitor patients remotely and send data
 - Cyber security measures are critical for protecting sensitive health information
 - What ethical considerations arise from collecting patient data